# Practice workbook · Calculate breathing rate or air volume from trace or table

Numeric work on breathing: breaths per minute, volume differences, minute volume and oxygen used.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-BREATHING-RATE-VOLUME-CALC

# 1. Question 2 · part b.iii · 3 marks

June 2023 Paper 1

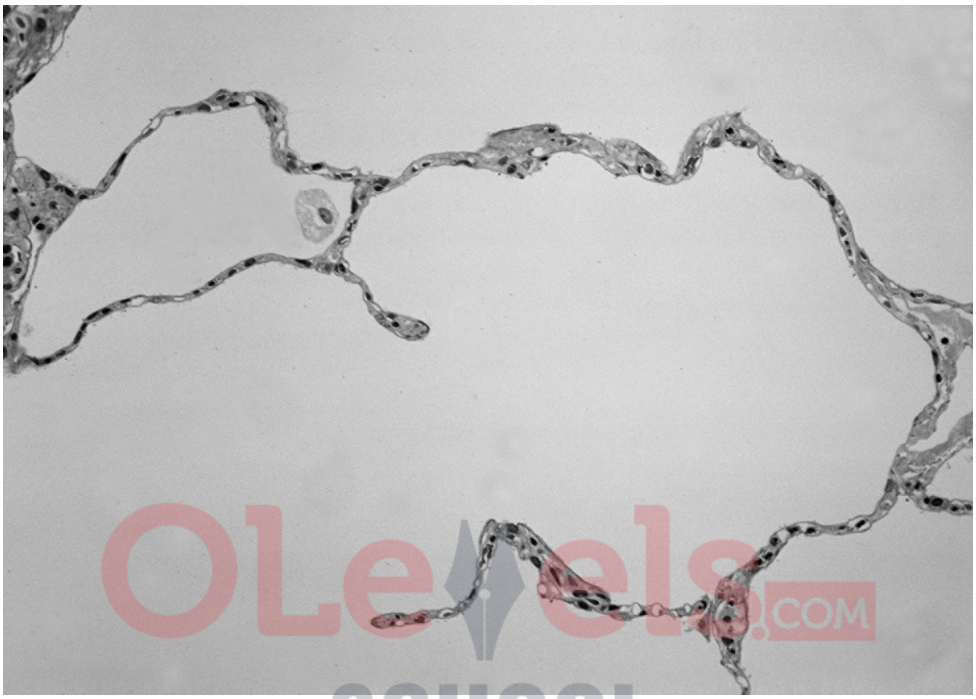
Worked-example retry · Attempt part b.iii only.

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**2** The photograph shows alveoli in the lungs. Alveoli are the site of exchange of gasses between the air and the blood.



(Source: © Ed Reschke/Getty Images)

(a) (i) State, in the correct order, the structures that air must pass through from the atmosphere to the alveoli.

**(2)**

.....

.....

(ii) State two features, shown in the diagram, that allow the efficient exchange of gases.

**(2)**

1 .....

.....

2 .....

.....

# 1. Question 2 · part b.iii · 3 marks

June 2023 Paper 1

Worked-example retry · Attempt part b.iii only.

(b) (i) Which is the percentage of carbon dioxide in the atmosphere?

**(1)**

- ☒ **A** 0.01
- ☒ **B** 0.04
- ☒ **C** 0.10
- ☒ **D** 0.50

(ii) The table shows the results obtained when a person at rest breathes in air that contains different percentages of carbon dioxide.

| **Percentage (%) carbon dioxide in air** | 0.02 | 1.60 | 3.00 | 6.00 |
|---|---|---|---|---|
| **Tidal volume in $cm^3$** | 500 | 700 | 1200 | 2100 |
| **Rate of breathing in breaths per minute** | 14 | 15 | 16 | 28 |

Calculate the volume of air in $dm^3$ breathed in per minute when the person breathes air containing 3.00% carbon dioxide.

[1 $dm^3$ = 1000 $cm^3$]

**(2)**

volume of air = ..... $dm^3$

(iii) The air a person breathes in contains 20% oxygen. The body uses 20% of this oxygen.

Calculate the volume of oxygen used per minute when the person breathes air containing 1.60% carbon dioxide.

**(3)**

volume of oxygen = ..... $cm^3$

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# 1. Question 2 · part b.iii · 3 marks

June 2023 Paper 1

Worked-example retry · Attempt part b.iii only.

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(iv) Explain the effect of increasing the percentage of carbon dioxide in the air a person breathes in on the tidal volume and rate of breathing.

**(4)**

**(Total for Question 2 = 14 marks)**

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Working and answer

# 2. Question 2 · part b.i · 2 marks

June 2021 Paper 2

Further practice · Attempt part b.i only.

**2** Aerobic respiration is the process that releases energy for use in the body.

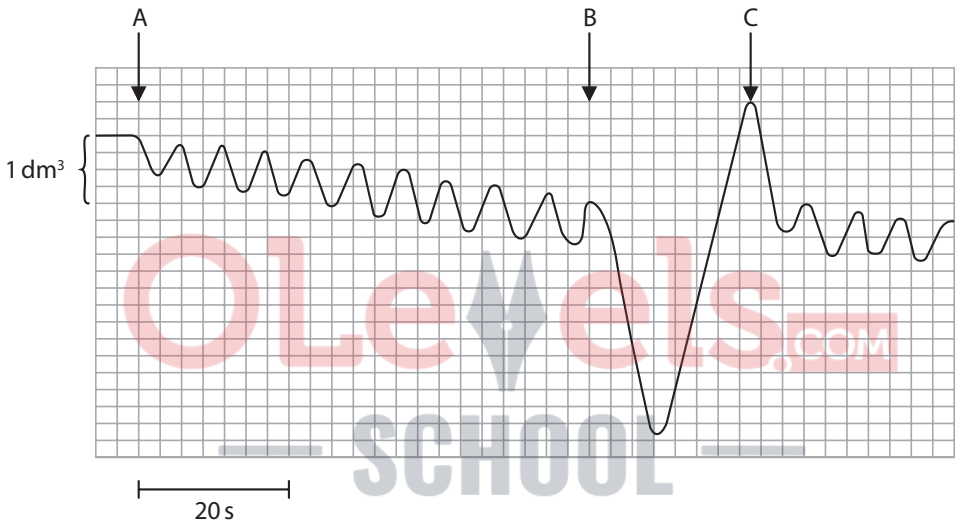
(a) Give the word equation for the process of aerobic respiration.

**(4)**

..... + ..... → ..... + ..... + energy

(b) A spirometer is used to measure the volumes of air breathed in and the volumes of air breathed out.

The spirometer trace shows the readings taken of a person at rest.




(i) Calculate the rate of breathing between points A and B shown on the trace.

**(2)**

rate = ..... breaths per minute

(ii) Calculate the difference in the volume of air breathed in and the volume of air breathed out between points B and C.

**(3)**

difference in volume = ..... $dm^3$

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P 6 6 4 3 3 A 0 4 2 4

## 2. Question 2 · part b.i · 2 marks · Working and answer

Working and answer

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# 3. Question 2 · part b.ii · 3 marks

June 2021 Paper 2

Further practice · Attempt part b.ii only.

**2** Aerobic respiration is the process that releases energy for use in the body.

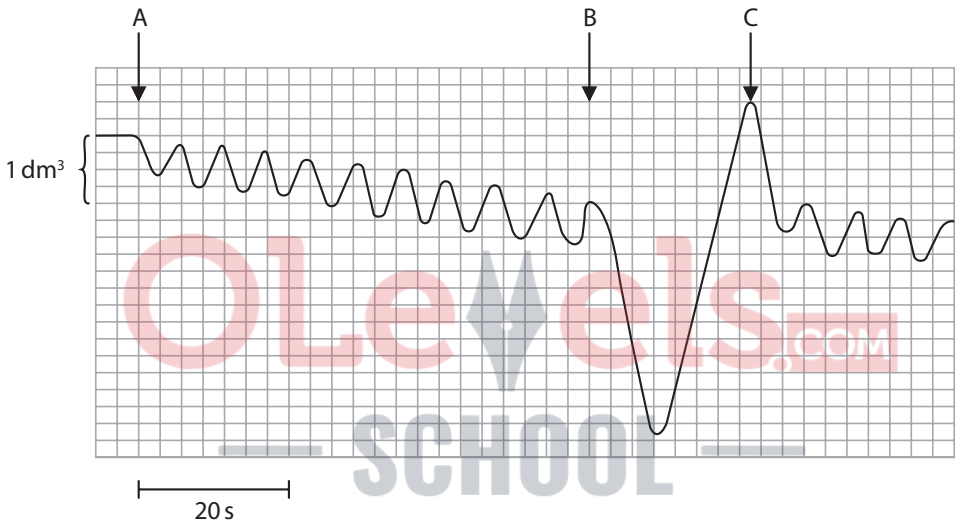
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## 3. Question 2 · part b.ii · 3 marks

June 2021 Paper 2

Further practice · Attempt part b.ii only.

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(c) Describe the differences that would be seen in the trace between points A and B if the person had been exercising vigorously before the readings were taken.

**(2)**

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**(Total for Question 2 = 11 marks)**

Working and answer

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# 4. Question 2 · part b.ii · 2 marks

June 2023 Paper 1

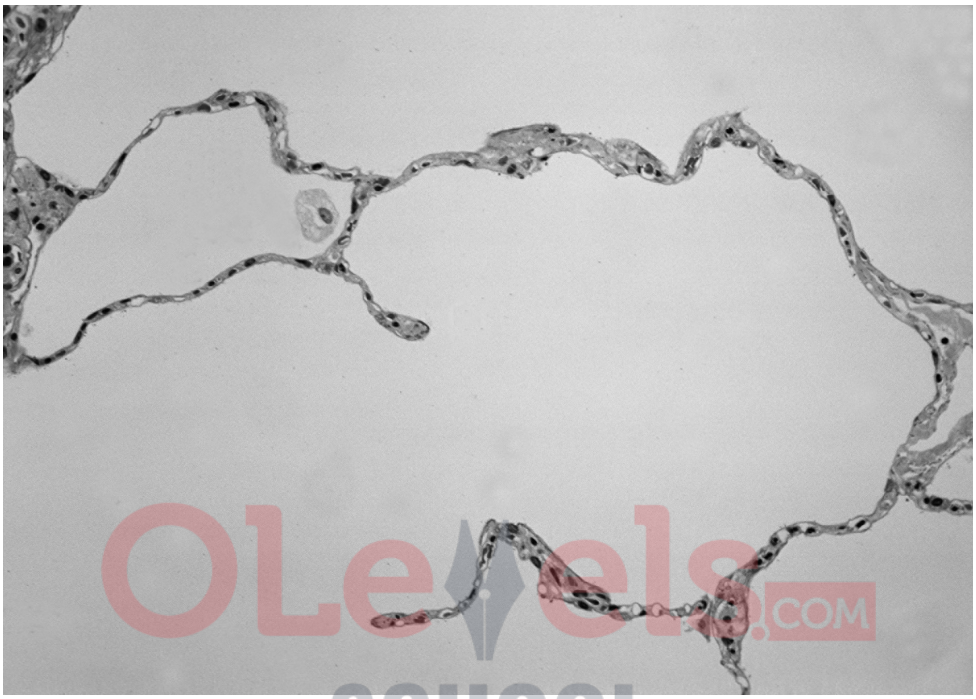
Further practice · Attempt part b.ii only.

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June 2023 Paper 1

Further practice · Attempt part b.ii only.

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P 7 2 5 8 5 A 0 4 2 0

## 4. Question 2 · part b.ii · 2 marks · Working and answer

Working and answer

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